Informatica® Cloud Application Integration
January 2025

Function Calling using Google Vertex AI

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Preface

Use *Function Calling using Google Vertex AI* to learn how to create the payload for Application Integration processes and execute them dynamically based on a user query. This guide assumes that you have an understanding of the Google Vertex AI Connector concepts.

CHAPTER 1

Introduction to Function Calling using Google Vertex AI recipe

The Function Calling using Google Vertex AI recipe is based on REST and SOAP APIs. The recipe allows you to create the payload for Application Integration processes and execute them dynamically based on a user query.

The recipe allows you to specify the user prompt and enter the function declaration in the JSON format to query other processes based on a function call response using a guide.

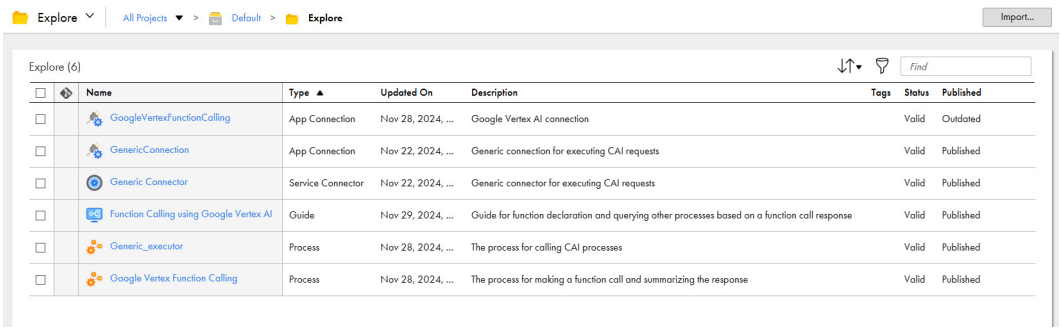
The process makes a request to Google Vertex AI with the specified function declaration and user prompt. Google Vertex AI recognizes the query, extracts key variables and values, and returns them in the appropriate format. Based on the received response, a function call is generated with function arguments and a link to the process execution.

The user confirms the accuracy of the function call response to make the call to Application Integration process or refuses to make function call again. The process requests the Application Integration recipe based on the function-calling response. The user then receives a message about the success of the recipe call and a response from the Application Integration recipe.

Function Calling using Google Vertex AI recipe contents

The Function Calling using Google Vertex AI recipe contains a service connector, app connections, a guide, and processes.

The following image shows the assets that the Function Calling using Google Vertex AI recipe package contains:



Function Calling using Google Vertex AI recipe assets

The following table lists the assets that the Function Calling using Google Vertex AI recipe package contains:

Asset Name	Asset Type	Description
Generic Connector	Service connector	Generic connector for executing an Application Integration request.
GoogleVertexFunctionCalling	App connection	Google Vertex AI connection.
GenericConnection	App connection	Generic connection for executing an Application Integration request.
Generic_executor	Process	Process to call another Application Integration processes.
Google Vertex Function Calling	Process	Process for making a function call and summarizing the response.
Function calling using Google Vertex AI	Guide	Guide for declaring a function and querying other processes based on a function call response.

CHAPTER 2

Using the Function Calling using Google Vertex AI recipe

To use the Function Calling using Google Vertex AI recipe, you must perform the following steps manually:

1. Copy and access the recipe.
2. Publish the Generic Connector connector.
3. Publish the GenericConnection connection.
4. Configure and publish the Google Vertex AI connection.
5. Configure and publish the processes.
6. Publish and run the guide.

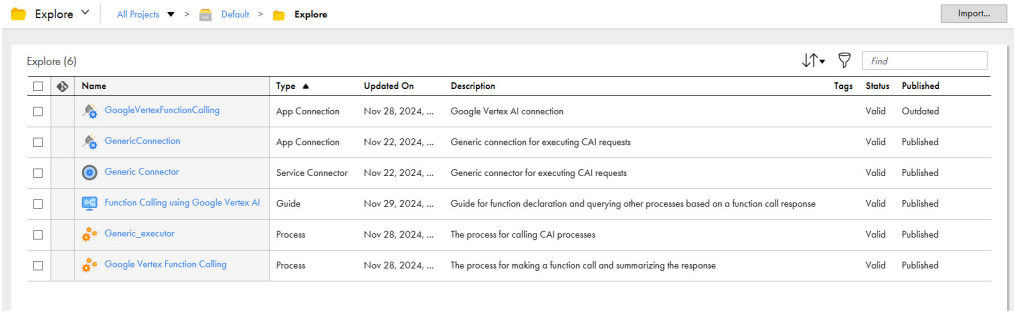
Step 1. Copy and access the recipe

Copy the pre-configured assets in the recipe to a separate project or folder.

1. Open the **Function Calling using Google Vertex AI** recipe and click **Use**.
2. Select the location where you want to copy the recipe, and then click **Continue**.
3. In the **Copying the recipe** dialog box, click **OK**.

   It might take some time for the recipe to get copied. You will receive a notification when the recipe is ready for use.
4. After the recipe is copied, click **Explore** to access the recipe content.
5. Navigate to the project or folder where you copied the recipe or enter the recipe name in the **Find** box.

   All the assets in the recipe are displayed as shown in the following image:



Step 2. Publish the Generic Connector service connector

To publish the **Generic Connector** service connector, open the **Generic Connector** service connector and click **Publish**.

Step 3. Publish the GenericConnection connection

To publish the **GenericConnection** connection, open the **GenericConnection** connection and click **Publish**.

Step 4. Configure and publish the Google Vertex AI connection

Configure the project ID, private key ID, private key, client email, and location to connect to Google Vertex AI, and then publish the GoogleVertexFunctionCalling connection.

1. Open the **GoogleVertexFunctionCalling** connection.
2. In the **Connection Properties** section, enter values for the following properties:

Property	Description
Project_ID	The project ID to generate a valid access token.
Location	The region to process the request.
Private_Key_ID	The private key ID associated with the Google Vertex AI service account.
Client_Email	The email address associated with the Google Vertex AI service account.
Private_Key	The Google Vertex AI private key associated with the service account. Enter the PKCS1 certificate as a Base64-encoded string in the following format: `-----BEGIN PRIVATE KEY----- ..... n-----END PRIVATE KEY-----`

3. Save and publish the connection.

Step 5. Configure and publish the processes

Configure the Google Vertex LLM model version and service URL, and publish the processes.

1. To publish the **Generic_executor** process, click **Actions** in the row that contains the process and select **Publish**.
2. Open the **Google Vertex Function Calling** process.

3. On the **Assignments** tab of the **Set Flow Configuration** step, enter values in the following fields:
   - In the **LLM_Model** field, enter the model ID of the model that you want to use. The model ID is set to gemini-1.5-pro-002, by default.
   - In the **Generation_Config** field, enter the instructions using the Expression Editor, as shown in the following sample code:

     ```
     <generationConfig>
         <temperature>0.2</temperature>
         <maxOutputTokens>1024</maxOutputTokens>
         <topP>0.8</topP>
     </generationConfig>
     ```

     For the **Generation_Config** field, enter values for the following properties:

Property	Description
temperature	Controls the randomness of the model's output. A lower value makes the output more deterministic, while a higher value increases randomness and creativity. For example, a temperature of 0.5 balances between deterministic and creative outputs.
maxOutputTokens	Defines the maximum number of tokens the model can generate in its response. Setting a limit ensures that the response is concise and fits within the desired length constraints.
topP	Determines the cumulative probability threshold for token selection. The model considers the smallest set of tokens whose cumulative probability meets or exceeds topP. For example, if topP is set to 0.1, the model considers only the top 10% most probable tokens at each step.

     To add additional generation config parameters, see the Generative AI on Vertex AI documentation.

4. On the **Assignments** tab of the **Prepare Default Function Declaration** step, enter the process URL of the recipe in the **Function_Declaration** field using the Expression Editor, as shown in the following sample code:

```
<tools>
    <functionDeclarations>
        <name>Update_Amazon_Bedrock_Knowledge_Base_executor</name>
        <description>This function processes a user-provided prompt to identify and
extract key information based on predefined parameters. For instance, it can detect
specific identifiers such as Knowledge_Base_ID or Email_Address by scanning the text
for patterns or matches associated with these keywords. The function returns the
extracted data for further processing or validation, facilitating automated
workflows where structured information is derived from unstructured input.</
description>
        <parameters>
            <type>object</type>
            <properties>
                <Process_URL>
                    <type>string</type>
                    <description>Fixed URL to execute process</description>
                    <enum><Service_URL_of_the_Application_Integration_Process></enum>
                </Process_URL>
                <Payload>
                    <type>array</type>
                    <description>Structured payload for processes</description>
                    <items>
                        <description>Fields in payload for processes</description>
                        <type>object</type>
                        <properties>
                            <Bucket_Name>
                                <type>string</type>
                                <description>The bucket name in AWS S3 is a unique
```

```
identifier for an S3 bucket.</description>
                            </Bucket_Name>
                            <Knowledge_Base_ID>
                                <type>string</type>
                                <description>The unique identifier of the knowledge
base. Sometimes, the knowledge base is abbreviated as KB.</description>
                            </Knowledge_Base_ID>
                            <Email_Address>
                                <type>string</type>
                                <description>Email address where the process result
will be sent.</description>
                            </Email_Address>
                            <Data_Source_ID>
                                <type>string</type>
                                <description>The unique identifier of the data
source. Sometimes, the data source is abbreviated as DS.</description>
                            </Data_Source_ID>
                        </properties>
                        <required>Knowledge_Base_ID</required>
                        <required>Email_Address</required>
                    </items>
                </Payload>
            </properties>
            <required>Process_URL</required>
            <required>Payload</required>
        </parameters>
    </functionDeclarations>
    <functionDeclarations>
        <name>Synchronize_ServiceNow_Incidents_with_Jira_Issues_executor</name>
        <description>This function processes a user-provided prompt to identify and
extract key information based on predefined parameters. For instance, it can detect
specific identifiers such as Jira_Issue_Type_ID or Jira_Project_ID by scanning the
text for patterns or matches associated with these keywords. The function returns
the extracted data for further processing or validation, facilitating automated
workflows where structured information is derived from unstructured input.</
description>
        <parameters>
            <type>object</type>
            <properties>
                <Process_URL>
                    <type>string</type>
                    <description>Fixed URL to execute process</description>
                    <enum><Service_URL_of_the_Application_Integration_Process></enum>
                </Process_URL>
                <Payload>
                    <type>array</type>
                    <description>Structured payload for processes</description>
                    <items>
                        <description>Fields in payload for processes</description>
                        <type>object</type>
                        <properties>
                            <Date_For_Search>
                                <type>string</type>
                                <description>Search date in the YYYY-MM-DD format.</
description>
                            </Date_For_Search>
                            <Filter>
                                <type>string</type>
                                <description>Name of the category to be filtered</
description>
                            </Filter>
                            <Email>
                                <type>string</type>
                                <description>Email address where the process result
will be sent.</description>
                            </Email>
                            <Jira_Issue_Type_ID>
                                <type>string</type>
                                <description>The unique identifier of Jira issue
```

```
type</description>
                            </Jira_Issue_Type_ID>
                            <Jira_Project_ID>
                                <type>string</type>
                                <description>The unique identifier of Jira project</
description>
                            </Jira_Project_ID>
                            <Sync_Assignee>
                                <type>boolean</type>
                                <description>Boolean variable whether to synchronize
the Assignee</description>
                            </Sync_Assignee>
                        </properties>
                        <required>Email</required>
                        <required>Jira_Issue_Type_ID</required>
                        <required>Jira_Project_ID</required>
                    </items>
                </Payload>
            </properties>
            <required>Process_URL</required>
            <required>Payload</required>
        </parameters>
    </functionDeclarations>
</tools>
```

This example applies to and can execute the **Synchronize_ServiceNow_Incidents_with_Jira_Issue** and **Update_Amazon_Bedrock_Knowledge_Base** recipes.

5. Save and publish the process.

Step 6. Publish and run the guide

You can declare functions and query other processes based on a function call response.

1. Open the **Function Calling using Google Vertex AI** guide.
2. On the **Start** tab of the Start step, ensure that the **Run As** field is set to **Current User**.
3. Save and publish the guide.
4. On the **Actions** menu, click **Run**. Alternatively, you can copy the execution URL from the **Properties Details** dialog box to run the guide.
5. On the **Instructions** page, enter the following values:
   - In the **Enter a prompt below** field, enter your question based on the function declaration.
   - Optionally, in the **Function declaration** field, enter the custom function declaration. By default, the function declaration value is set for the Synchronize ServiceNow Incidents with Jira Issues recipe and Update Amazon Bedrock Knowledge Base recipe in the **Prepare Default Function Declaration** step of the **Google Vertex Function Calling** process.
   - Select the function calling confirmation as **No** if you don't want to receive a summary of the function call response and omit the confirmation step before the function is executed. By default, the function calling confirmation is selected as **Yes**.
6. Click **Continue**.
   You will receive a summary of the function call request in response if the function call is generated. Otherwise, an error message appears.
7. Click **Confirm** to confirm the summary of the function call response and make a call to the CAI process or click **Try again**.
   You will receive a response that contains the process call status, the process execution status, and the process response.

Note: If the process you are calling returns any response, you will see the response on the Answer screen. However, if the process runs for more than 59 seconds, you will see a message about the successful process call, but the result can be viewed in Application Integration Console.

You can also use the embed code to embed the guide into an HTML document of a third-party application.

Test Function Calling using Google Vertex AI

You can test the Function Calling using Google Vertex AI recipe with the function declaration value that is set as default in the **Prepare Default Function Declaration** step of the **Google Vertex Function Calling** process.

The default value applies to and can execute the Synchronize ServiceNow Incidents with Jira Issues recipe and the Update Amazon Bedrock Knowledge Base recipe.

For example, in the **Function Calling using Google Vertex AI** guide, you can enter one of the following prompts for the correct function call for these recipes:

- *Please make an ingestion job for KB ON1FS86KFX with DS REQJVGUX8S and send the result to example@gmail.com*
- *Synchronize my incidents in the Software category for November 19, 2024, project id 1001, Jira issue type id 1001, you also need to synchronize the Assignee and send the result to example@gmail.com*

The function name must start with a letter or an underscore, and can contain alphanumeric characters, underscores (_), periods (.), and hyphens (-). For more information, see Google Vertex AI documentation.

The structure of the **Function_Call_Answer** is important for further execution of the request to the Application Integration process. To set your **Function_Declaration** in the guide, you must specify it in the JSON format as shown in the following example:

```
{
    "tools": [{
            "functionDeclarations": [{
                    "name": "Update_Amazon_Bedrock_Knowledge_Base_executor",
                    "description": "This function processes a user-provided prompt to
identify and extract key information based on predefined parameters. For instance, it
can detect specific identifiers such as Knowledge_Base_ID or Email_Address by scanning
the text for patterns or matches associated with these keywords. The function returns
the extracted data for further processing or validation, facilitating automated
workflows where structured information is derived from unstructured input.",
                    "parameters": {
                        "type": "object",
                        "properties": {
                            "Process_URL": {
                                "type": "string",
                                "description": "Fixed URL to execution process",
                              "enum":
["<Service_URL_of_the_Application_Integration_Process>"]
                            },
                            "Payload": {
                                "type": "array",
                                "description": "Structured payload for processes",
                                "items": {
                                    "description": "Fields in payload for processes",
                                    "type": "object",
                                    "properties": {
                                        "Bucket_Name": {
                                            "type": "string",
                                            "description": "The bucket name in AWS S3 is
a unique identifier for an S3 bucket."
                                        },
```

```
                                        "Knowledge_Base_ID": {
                                            "type": "string",
                                            "description": "The unique identifier of the
knowledge base. Sometimes, the knowledge base is abbreviated as KB."
                                        },
                                        "Email_Address": {
                                            "type": "string",
                                            "description": "Email address where the
process result will be sent."
                                        },
                                        "Data_Source_ID": {
                                            "type": "string",
                                            "description": "The unique identifier of the
data source. Sometimes, the data source is abbreviated as DS."
                                        }
                                    },
                                    "required": ["Knowledge_Base_ID", "Email_Address"]
                                }
                            }
                        },
                        "required": ["Process_URL", "Payload"]
                    }
                }, {
                    "name": "Synchronize_ServiceNow_Incidents_with_Jira_Issues_executor",
                    "description": "This function processes a user-provided prompt to
identify and extract key information based on predefined parameters. For instance, it
can detect specific identifiers such as Jira_Issue_Type_ID or Jira_Project_ID by
scanning the text for patterns or matches associated with these keywords. The function
returns the extracted data for further processing or validation, facilitating automated
workflows where structured information is derived from unstructured input.",
                    "parameters": {
                        "type": "object",
                        "properties": {
                            "Process_URL": {
                                "type": "string",
                              "description": "Fixed URL to execution process",
                              "enum":
["<Service_URL_of_the_Application_Integration_Process>"]
                            },
                            "Payload": {
                                "type": "array",
                                "description": "Structured payload for processes",
                                "items": {
                                    "description": "Fields in payload for processes",
                                    "type": "object",
                                    "properties": {
                                        "Date_For_Search": {
                                            "type": "string",
                                            "description": "Search date in the YYYY-MM-
DD format."
                                        },
                                        "Filter": {
                                            "type": "string",
                                            "description": "Name of the category to be
filtered"
                                        },
                                        "Email": {
                                            "type": "string",
                                            "description": "Email address where the
process result will be sent."
                                        },
                                        "Jira_Issue_Type_ID": {
                                            "type": "string",
                                            "description": "The unique identifier of
Jira issue type"
                                        },
                                        "Jira_Project_ID": {
                                            "type": "string",
                                            "description": "The unique identifier of
Jira project"
                                        },
```

```
                                        "Sync_Assignee": {
                                            "type": "boolean",
                                            "description": "Boolean variable whether to
synchronize the Assignee"
                                        }
                                    },
                                    "required": ["Email", "Jira_Issue_Type_ID",
"Jira_Project_ID"]
                                }
                            }
                        },
                        "required": ["Process_URL", "Payload"]
                    }
                }

            ]

        }

    ]
}
```

For testing, you can construct another structure of a Function Declaration as required.